THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 6 · Term 1

Student edition

Outcomes: MA-6.1 sets, Venn diagrams and integers; MA-6.2 ratio, rate and percentages.

Lesson notes

MA-6.1 Sets (Weeks 1–4)

Notation { }, ∈, ∉; finite, infinite, empty, singleton and universal sets; union and intersection on Venn diagrams with class data (who plays cricket, who plays football). [picture: Venn diagram]

MA-6.1 Integers (Weeks 5–8)

Number line with negatives (temperatures in Skardu in winter); absolute value; ordering; the four operations with sign rules (same signs give +, different signs give − for × and ÷).

MA-6.2 Ratio, rate and percentages (Weeks 9–12)

Simplify ratios; share an amount in a ratio; continued ratios (a:b:c); unit rates (km/h, Rs per kg); one quantity as a percentage of another; percentage increase and decrease.

Workbook pages

Page	Outcome	Activity
1–2	MA-6.1	Set notation; Venn diagrams from a class survey.
3–4	MA-6.1	Integer number lines and sign-rule practice.
5–7	MA-6.2	Sharing in ratios; best-buy rates; percentage change.

Question bank

#	Outcome	Question	Your answer
1	MA-6.1	A = {1, 2, 3, 4}, B = {3, 4, 5}. A ∩ B =	{1, 2} / {3, 4} / {1, 2, 3, 4, 5} / { }
2	MA-6.1	A ∪ B for the same sets =	{3, 4} / {1, 2, 3, 4, 5} / {5} / {1, 2}
3	MA-6.1	The set of even numbers between 1 and 3 is a singleton set. True or false.	True / False
4	MA-6.1	−7 + 3 =	−10 / −4 / 4 / 10
5	MA-6.1	−5 − (−8) =	−13 / −3 / 3 / 13
6	MA-6.1	(−6) × (−4) =	−24 / 24 / −10 / 10
7	MA-6.1	−20 ÷ 4 =	5 / −5 / −16 / 80
8	MA-6.1	¦−9¦ (absolute value) =	−9 / 9 / 0 / 81

#	Outcome	Question	Your answer
9	MA-6.2	Simplify 12 : 18.	6 : 9 / 2 : 3 / 3 : 2 / 1 : 6
10	MA-6.2	Share Rs 600 in the ratio 2 : 3.	_____
11	MA-6.2	A car travels 150 km in 3 hours. Its rate is …	45 km/h / 50 km/h / 153 km/h / 450 km/h
12	MA-6.2	Increase 250 by 20%.	270 / 300 / 50 / 200
13	MA-6.2	15 out of 60 as a percentage =	15% / 25% / 40% / 4%